VIRTUAL MEMORY SIMULATOR

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PROJECT OVERVIEW

Algorithms Used:
FIFO
LRU
OPTIMAL
LFU
MFU
Second Chance
Enhanced Second Chance
Add Reference Bits

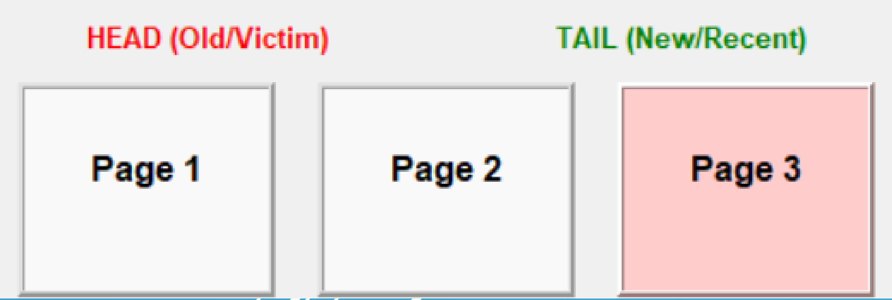
Patterns observed:
Belady Anamoly
High Locality
Thrashing
R/W for ESC
Timer Interrupt - ARB

Tech Stack:
Python
Libraries Used:
tkinter
random
collections
math

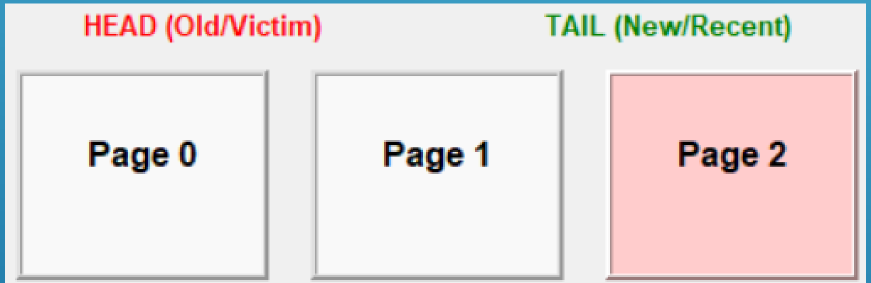
ALGORITHMS USED

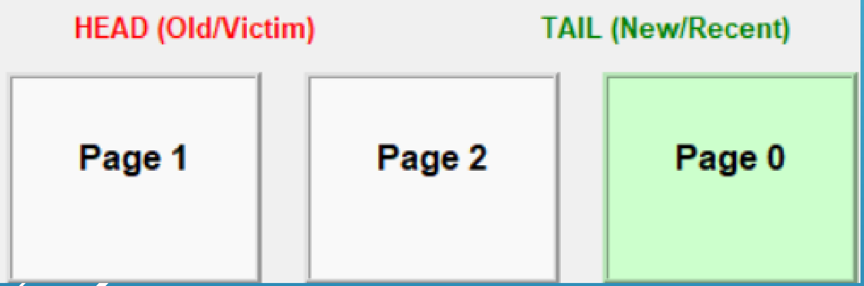
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

FIFO

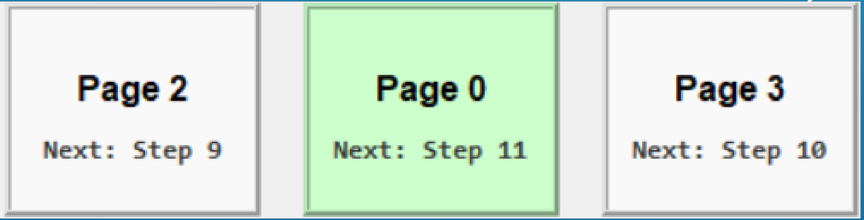


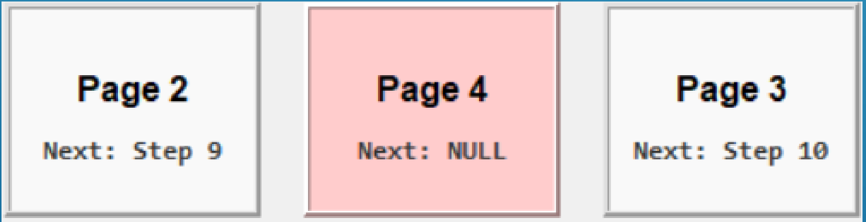
LRU





OPTIMAL





ALGORITHMS USED

9, 3, 2, 3, 3, 2,0,

LFU

Page 9 Uses: 1	Page 3 Uses: 3	Page 2 Uses: 2	Page 0 Uses: 1	Page 3 Uses: 3	Page 2 Uses: 2

MFU

Page 9 Uses: 1	Page 3 Uses: 3	Page 2 Uses: 2	Page 9 Uses: 1	Page 0 Uses: 1	Page 2 Uses: 2

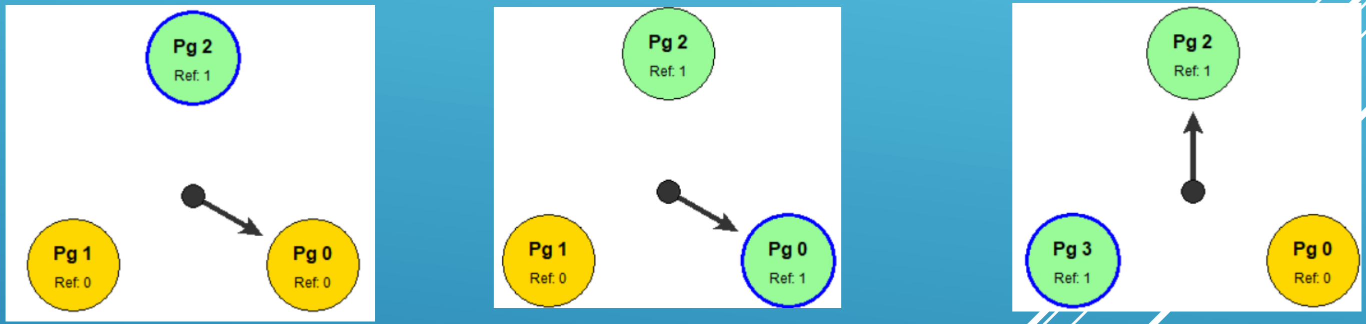
FIFO and LRU implemented through queue, though for LRU on hit, the page is again reupdated to other end while in FIFO , it does not change position. Optimal keeps a track of next occurrence of page call and removes the page which will come after longest time with NULL as infinity. LFU and MFU keep track of number of occurrences in cache and update based on it.

ALGORITHMS USED

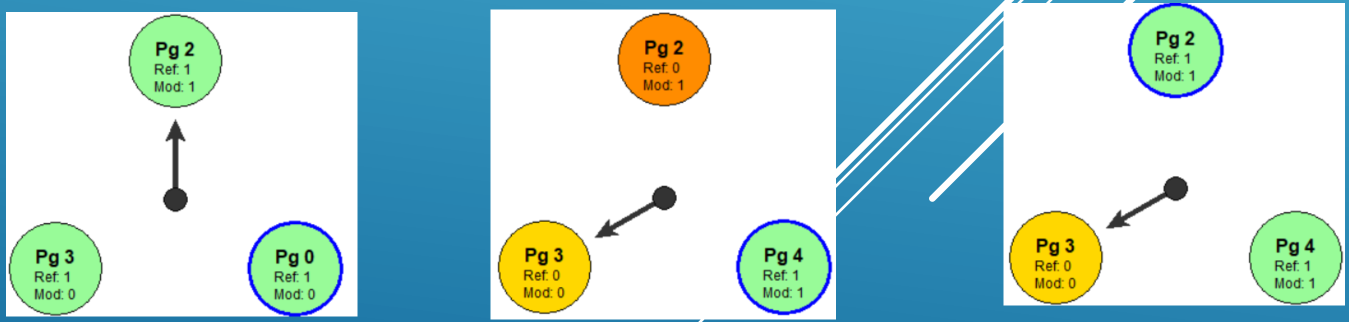
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

R, R, R, W, R, R, R, W, R, R, R, W, R, R, R, R, W, R, R, W

SC



ESC



Second Chance uses Reference Bit while Enhanced second chance uses Modification bit for R/W operations. Pointer moves to next page after putting a page in memory.

ALGORITHMS USED

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Additional Reference Bits

Page ID	Ref Bit	History Register (8-bit)
Page 1	1	11000000
Page 0	1	10000000
Page 7	1	00000000

Additional Reference Bits uses a Reference Bit and History register. History Register is updated at time of Page faults and Timer Interrupts which has been enabled for simulation. Register with lowest value is replaced during Page fault.

BELADY ANOMALY

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Algo | Frames	1	2	3	4	5	6	7	8
FIFO	12	12	9	10	5	5	5	5
LRU	12	12	10	8	5	5	5	5
OPTIMAL	12	9	7	6	5	5	5	5
LFU	12	12	10	8	5	5	5	5
MFU	12	12	9	10	5	5	5	5
Second Chance	12	12	9	10	5	5	5	5
ADD Ref Bits	12	12	9	10	5	5	5	5

CONCLUSION

We have successfully made a simulator for running multiple algorithms on paging algorithms. These include basic First-In-First-Out (FIFO), Least Recently Used (LRU), Optimal, Least Frequently Used, Most Frequently Used, Second Chance, Enhanced Second Chance and Adding Reference Bits. We have implemented correct data structures and have implemented animations for there visual understanding along with hit rate. We have also made graphs to see how each algorithm works given different number of frames and has helped in observing Belady's Anomaly.